NANYANG TECHNOLOGICAL UNIVERSITY

SC2107/CE2107
Microprocessor System Design and Development

The Analogue Interface

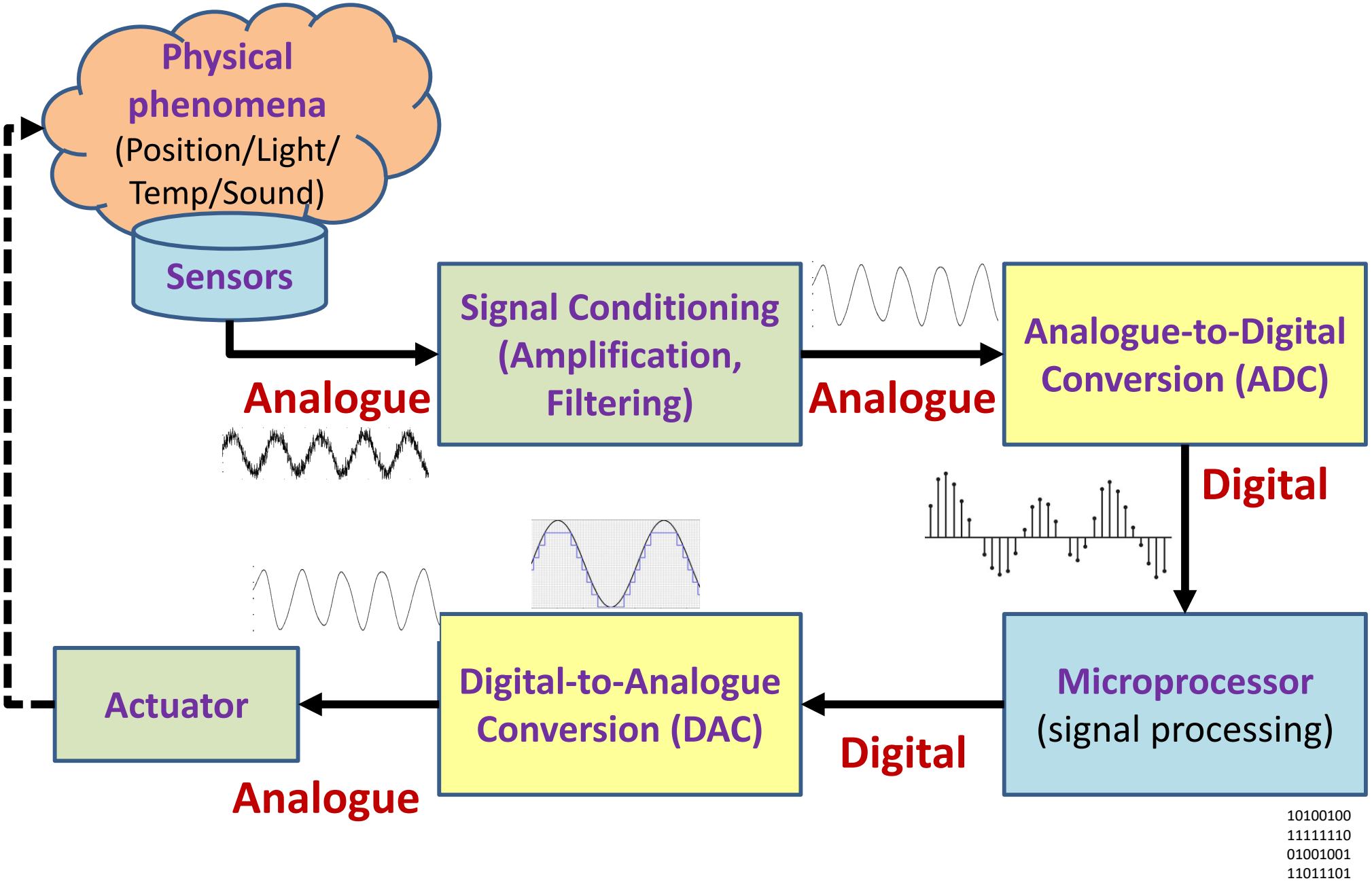
Topics Covered

- Signal Acquisition
- Digital-to-Analogue Conversion (DAC)
- Analogue to Digital Conversion (ADC)

Reference: Alan Clements, "Principles of Computer Hardware", 4th Edition, Oxford University Press (Section 11.6)

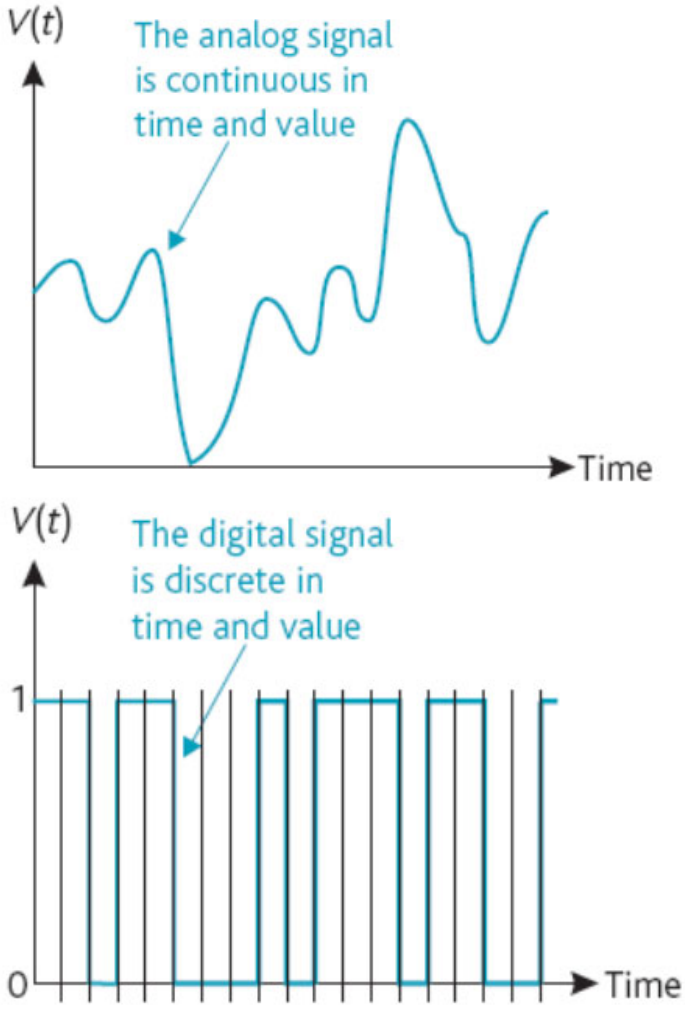
Signal Acquisition

The Measurement and Control Loop

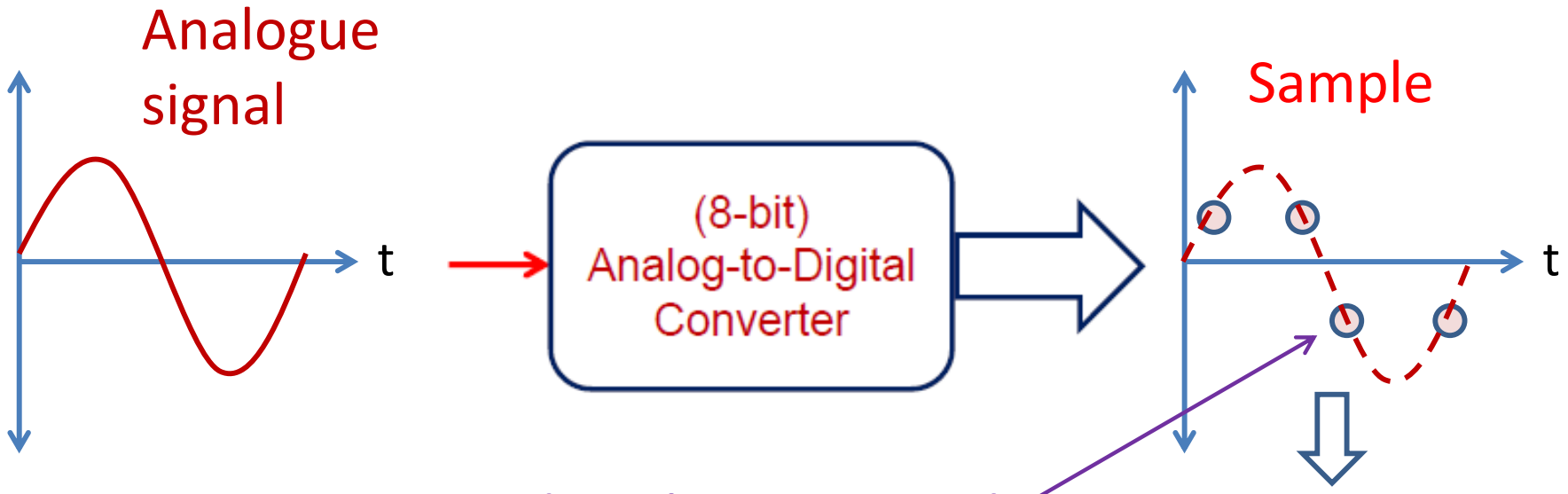


Analogue vs. Digital Signals

- When microprocessor is used to interact with real world signals (which are mainly analog in nature),
  - Converters such as Analog-to-Digital Converter (**ADC**) and Digital-to-Analog Converter (**DAC**) are needed
- Analog Signal
  - Continuous in time and amplitude
  - i.e. **infinite resolution**
- Digital Signal
  - Signal valid at discrete time only (due to sampling)
  - Amplitude valid at certain discrete levels only
  - i.e. **finite amplitude** and **time resolution**



Analogue to Digital Conversion



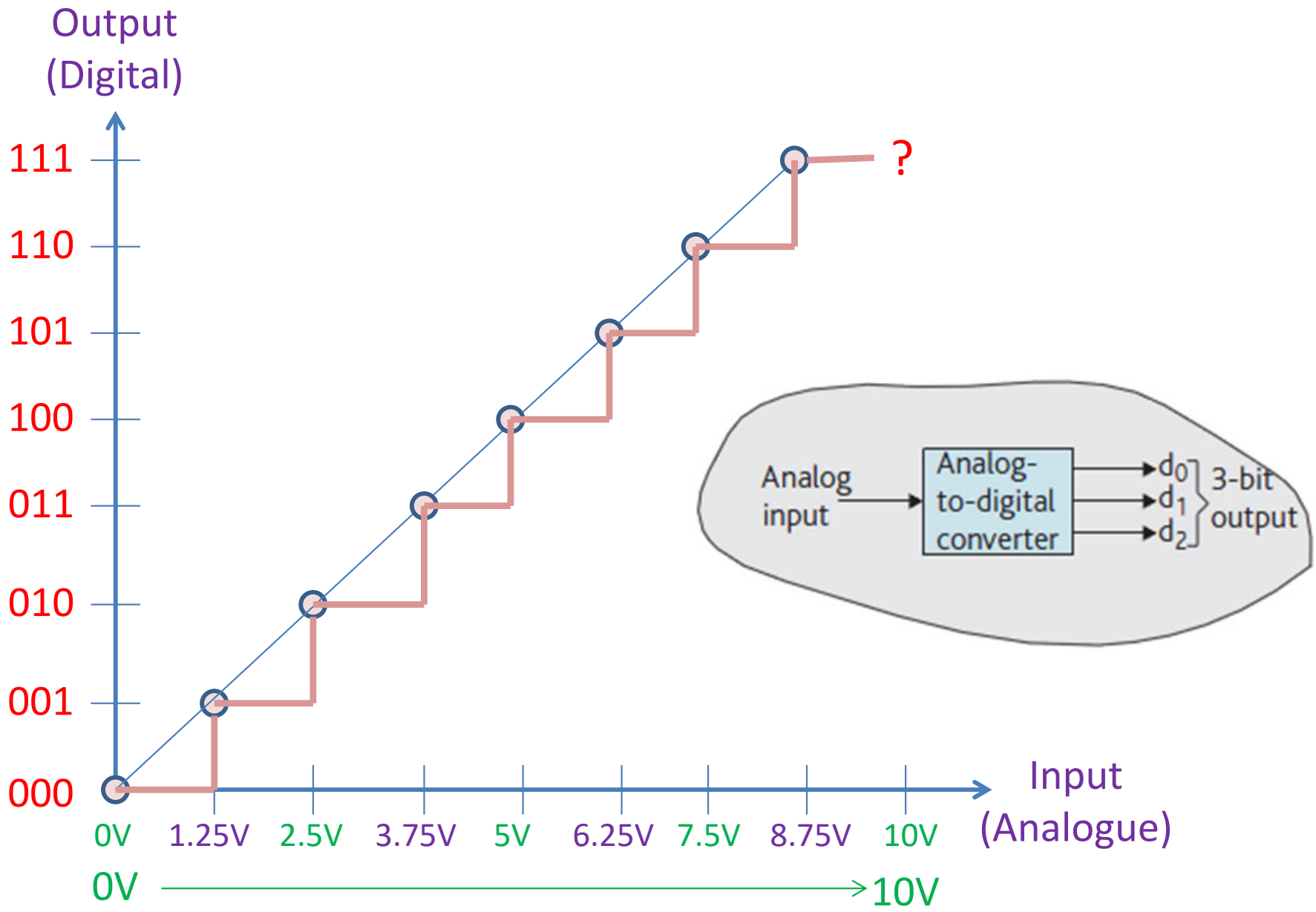
- Two operations:
  - Extract sample value of analogue signal
  - Convert sample value to digital value (in binary)
- Hence, there is loss of signal's detail when it is converted from analog to digital

Signal Quantization

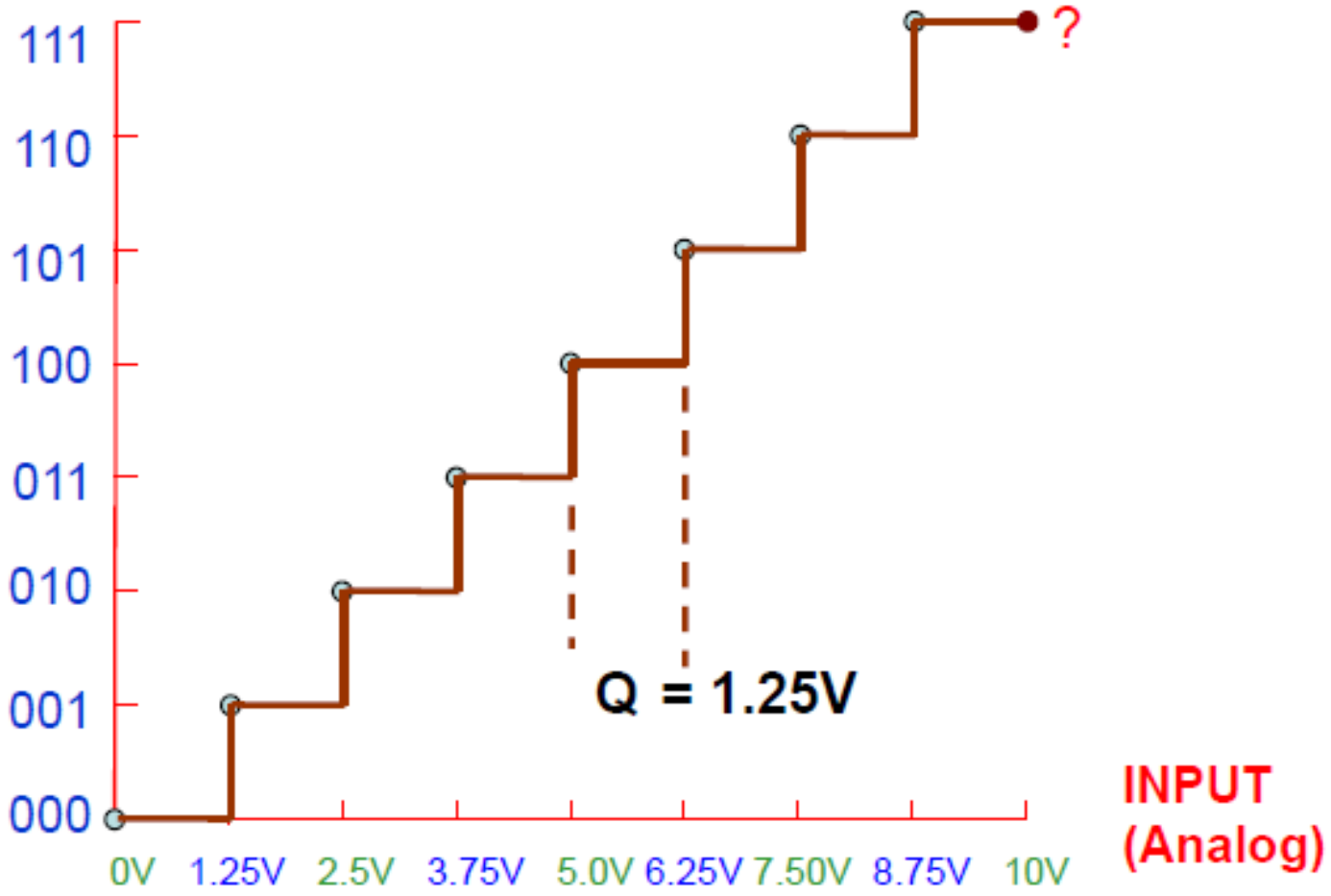
- Analogue to digital conversion requires quantization of continuous signal into multilevel signals
  - Results in loss of signal resolution
- Two fundamental questions:
  - How often should the conversion (sampling) is carried out?
  - How many levels (digital values) should the analogue input be divided? For example,
    - Input voltage range is 0V to 10V (can have any values in this range e.g. 1 mV, 1 µV, etc.) – Infinite analogue values
    - Output values of a 3-bit ADC is only 0, 1, 2, .., 7 (8 digital values)

Input-Output Characteristics of an ADC

Example: A 3-bit ADC Input-Output Characteristics



Quantization Q



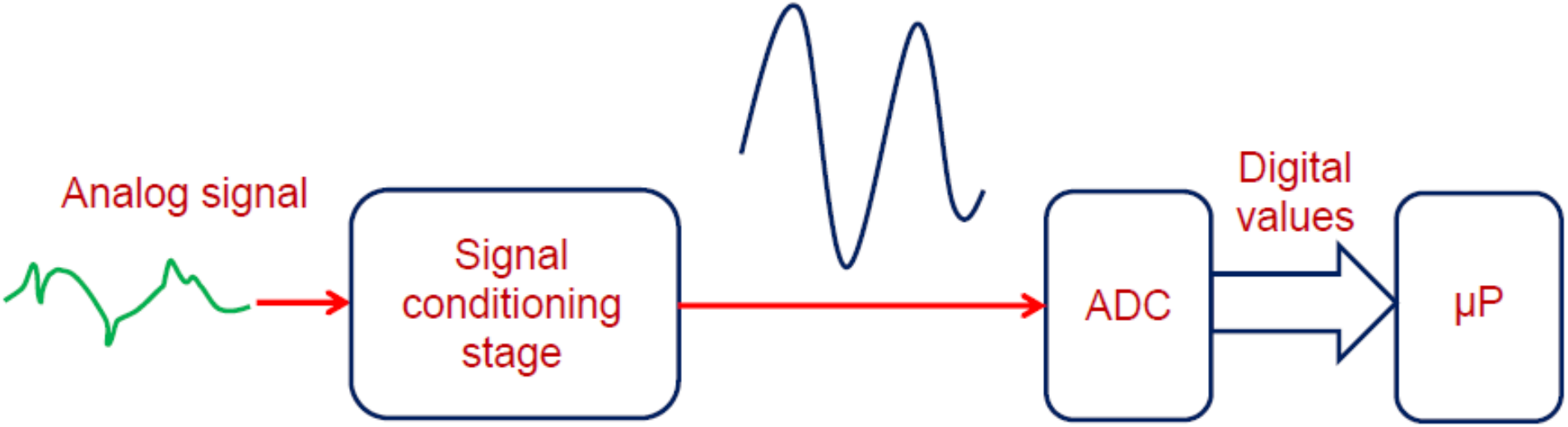
- Quantization Size (Quantum) $Q = FS/2^N = 10/2^3 = 1.25V$
  - The input voltage range for which its digital output value remains the same

Resolution of ADC

- Resolution of ADC can be expressed/described in term of
  - Number of bits (N) of ADC (e.g. 8-bit, 10-bit)
  - Actual Quantization size Q = $FS/2^N$
    - Where FS = Full Scale input
    - e.g. 1.25V (for a 0V-10V, 3-bit ADC)
    - 2.44mV (for a ±5V, 12-bit ADC)

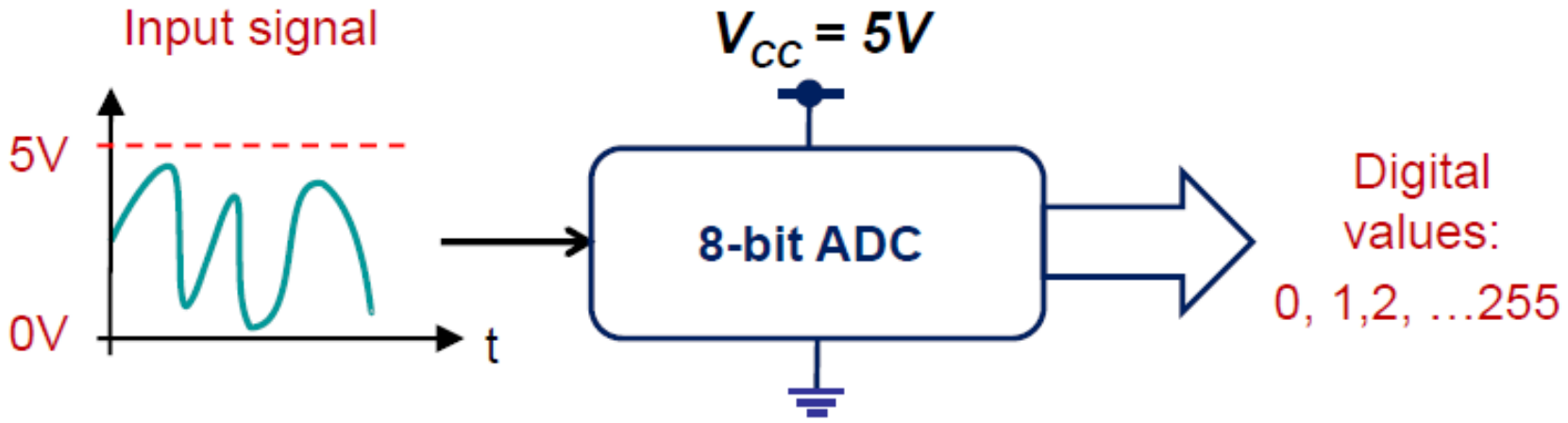
- Since $FS/2^N$ = Least Significant Bit size
  - Q is also described as 1 LSB

Input Signal Requirement



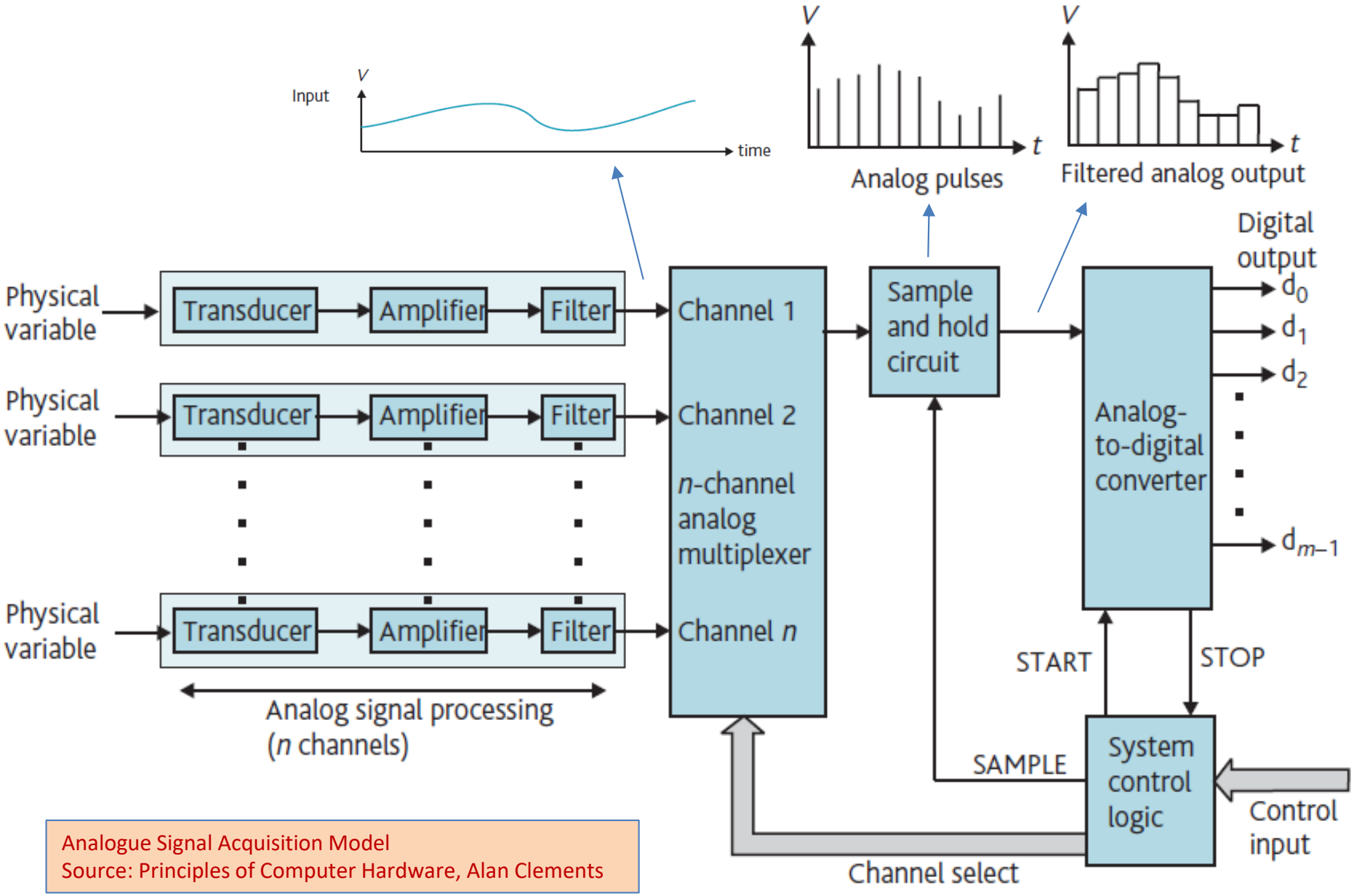
- What should be the signal characteristic at the input to the ADC (i.e. output of the signal conditioning stage)?
  - Amplitude should have a maximum level close to the Full Scale (FS) input to provide highest digital resolution
  - Noise level should be less than the quantization size (Q) or LSB such that it doesn't affect the signal value

Example



- Power supply of ADC ≈ FS input of the ADC
  - Hence, max input signal ≤ 5V
- Quantization size (Q) of 8-bit ADC = 5V/256 = 19.5mV
  - Hence, noise at input must be < 19.5mV

An Analogue Signal Acquisition Model



Analogue Signal Acquisition Model
Source: Principles of Computer Hardware, Alan Clements

Sampling and Aliasing

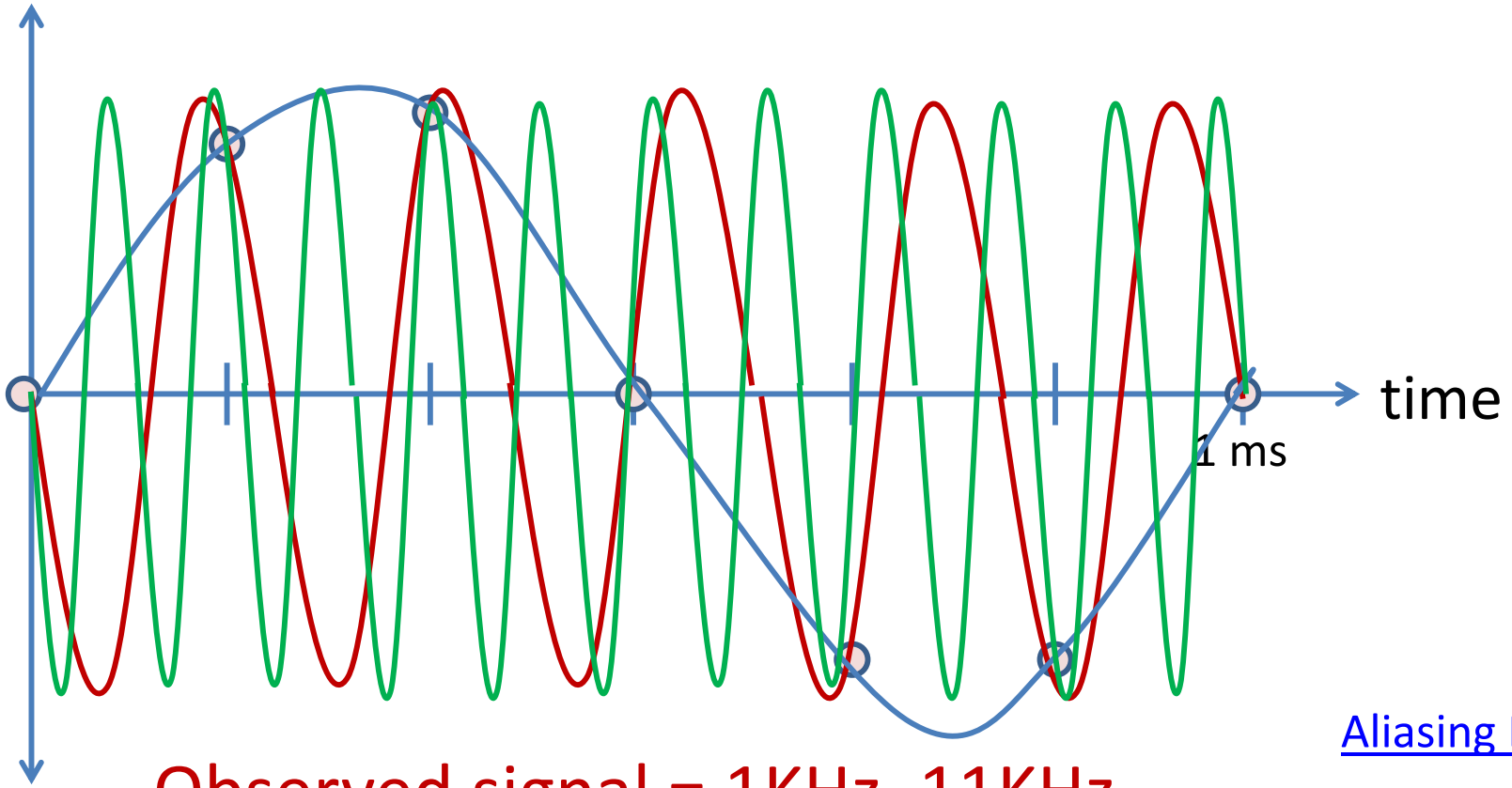
- Another important consideration when applying ADC is the sampling frequency used in the system
  - i.e. the number of samples of the signal taken over one second duration
- Intuitively, the more frequently we sample the signal
  - The more accurate is the digital values we obtained
- Correspondingly, if the number of samples taken is too little
  - The less precise is the information we captured
- In Analogue to Digital conversion, the sampling frequency must be **more then two times** of the highest frequency component in the signal (Nyquist rate)
  - otherwise **aliasing** effect will occur and information is lost

Aliasing

- Reconstruction from acquired signal samples introduces distortion if sampling rate is smaller than Nyquist rate

Signal = 5KHz (5 cycles in 1ms)

Sampling frequency = 6KHz (6 samples in 1 ms)



Aliasing Example

Observed signal = 1KHz, 11KHz, ….

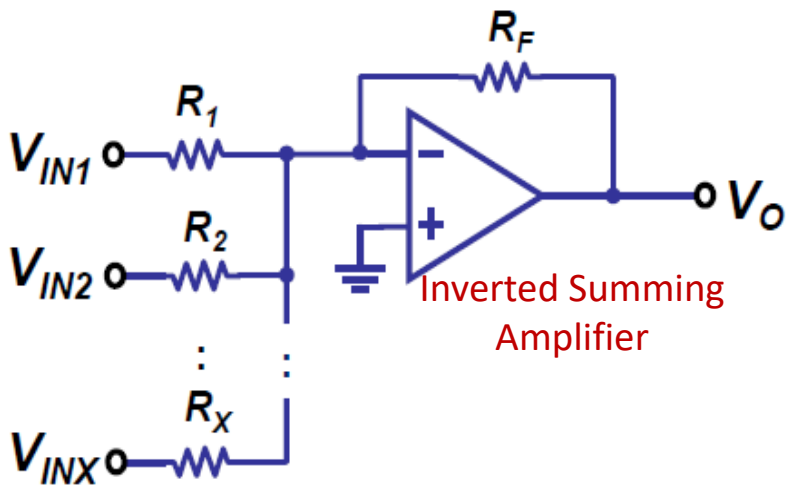
DAC and ADC

Digital to Analogue Conversion

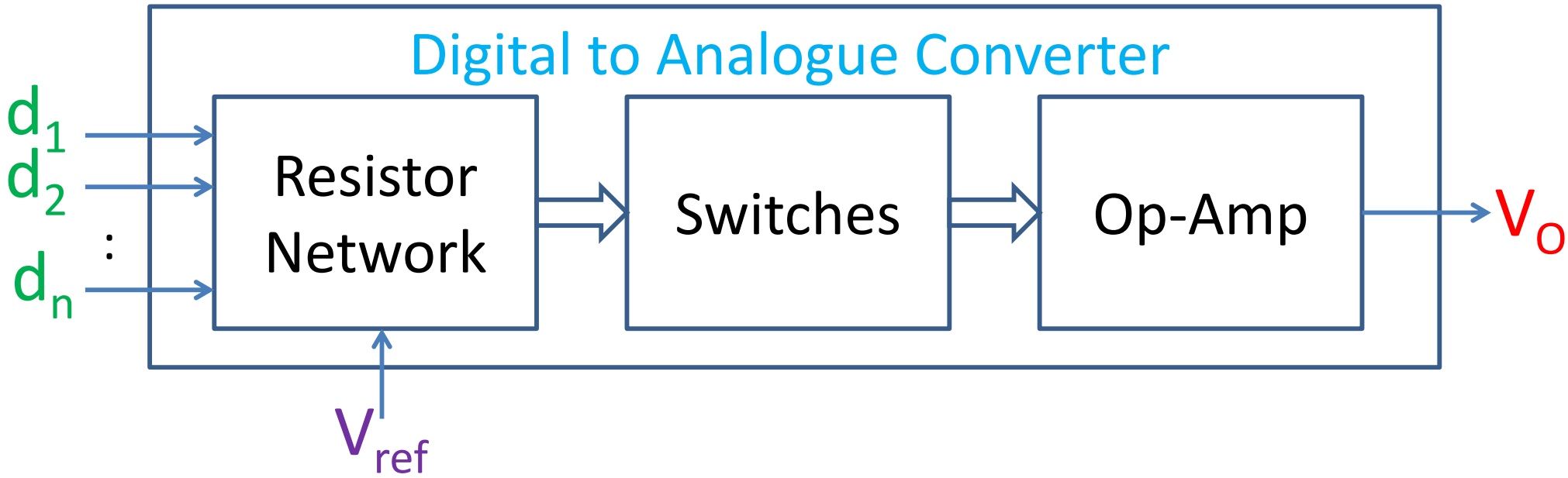
- Real world signals are more easily processed, stored and transferred in digital format
  - but need to eventually converted back to human recognizable analogue form
  - converted using Digital-to-Analogue converter (DAC)
- Examples:
  - Generation of audio signals in music players
    - MP3 files
    - Compact Disc
  - Generation of video signals in video player
    - DVD player
    - Video files replay on smart-phone/tablet

DAC Building Blocks

- DAC converts a binary value to analogue form
- This is achieved by:
  - Generating an analogue value that is proportional to each bit of the digital word
  - Adding the analogue values to give a composite analogue sum



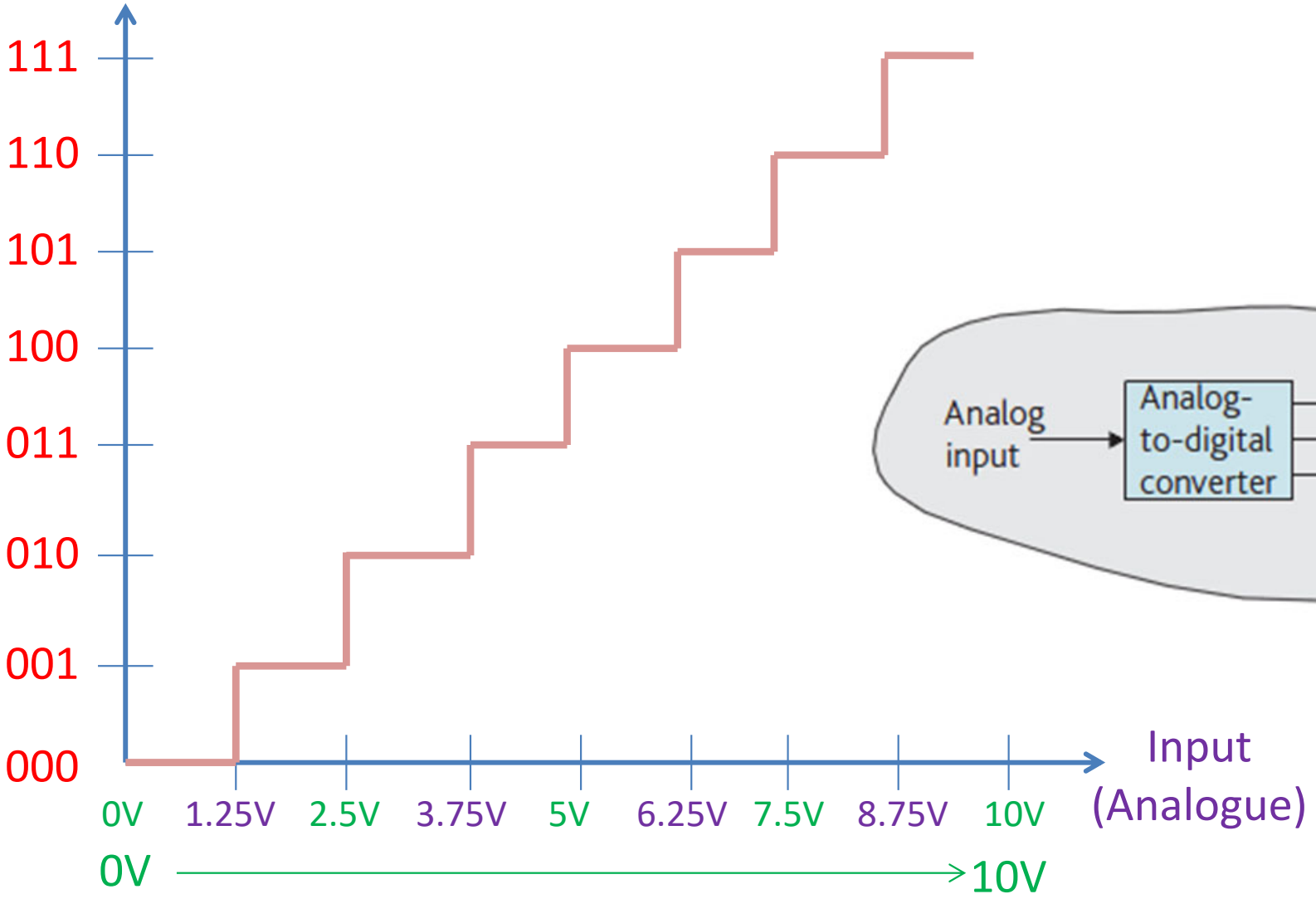
$V_O \approx K \bullet D$ $\quad D = d_1 2^{-1} + d_2 2^{-2} + \ldots + d_n 2^{-n}$



Types of ADC

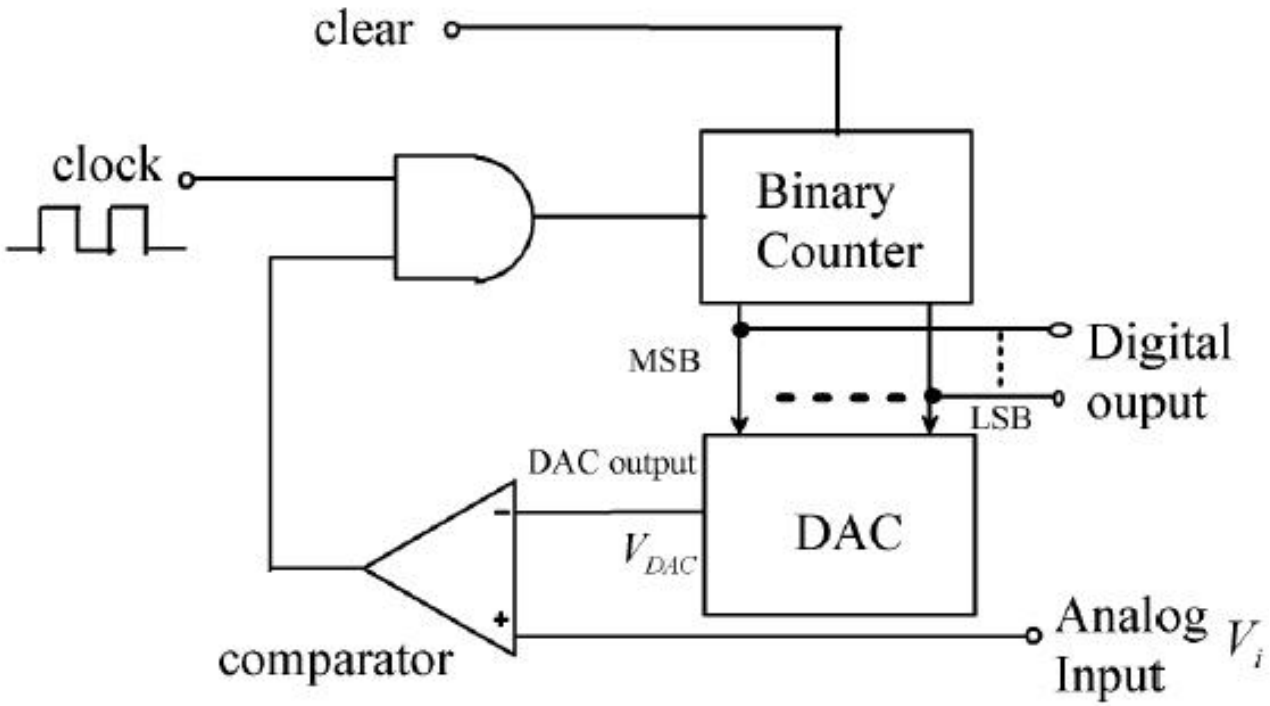
- Some types of ADC available in the market
  - Flash (Parallel)
  - Digital Ramp
  - Tracking
  - Successive Approximation
  - Integrating
  - Sigma-Delta
- Each type operates based on different principle
  - Different performance parameters (e.g. conversion time)
  - Cost effectiveness
  - Different application requirements

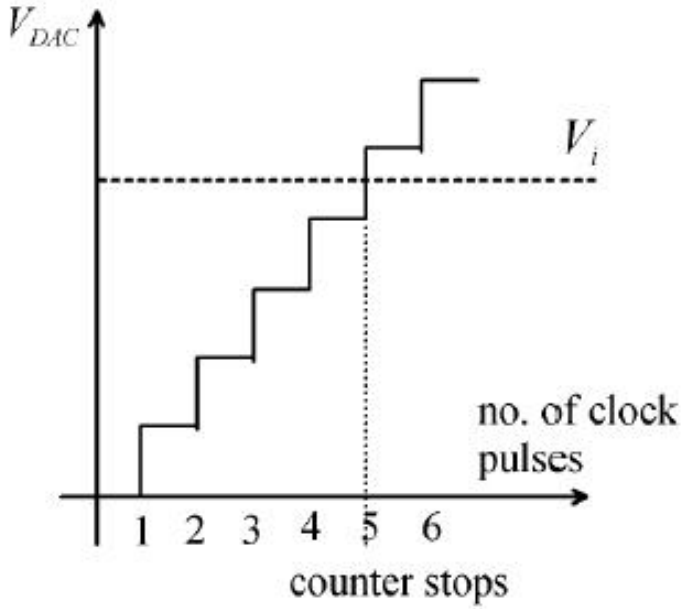
Input-Output Characteristic of an ADC



Digital Ramp ADC

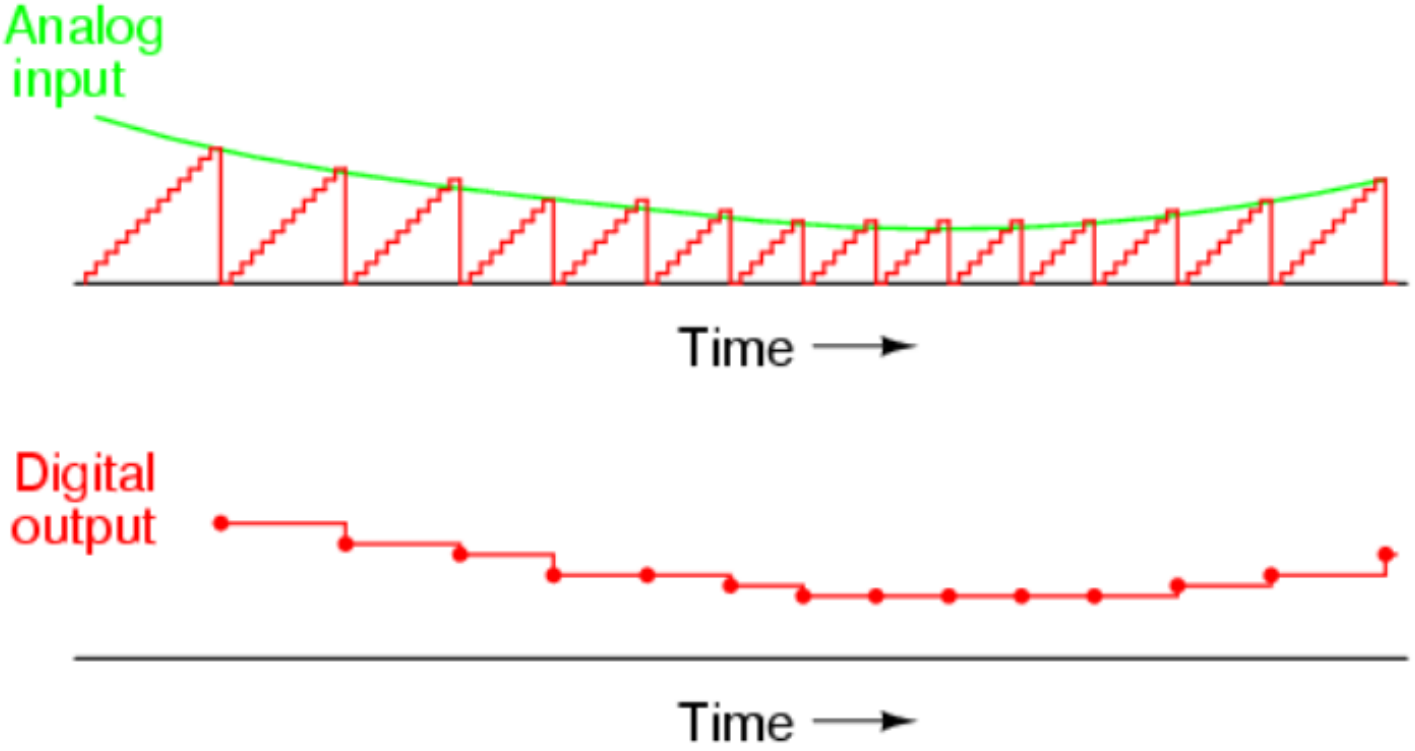
- Connect the output of a free-running binary counter to the input of a DAC
- Compare the analogue output of the DAC with the analogue input signal V_{in}
- Use the comparator's output to tell the counter when to stop counting and reset



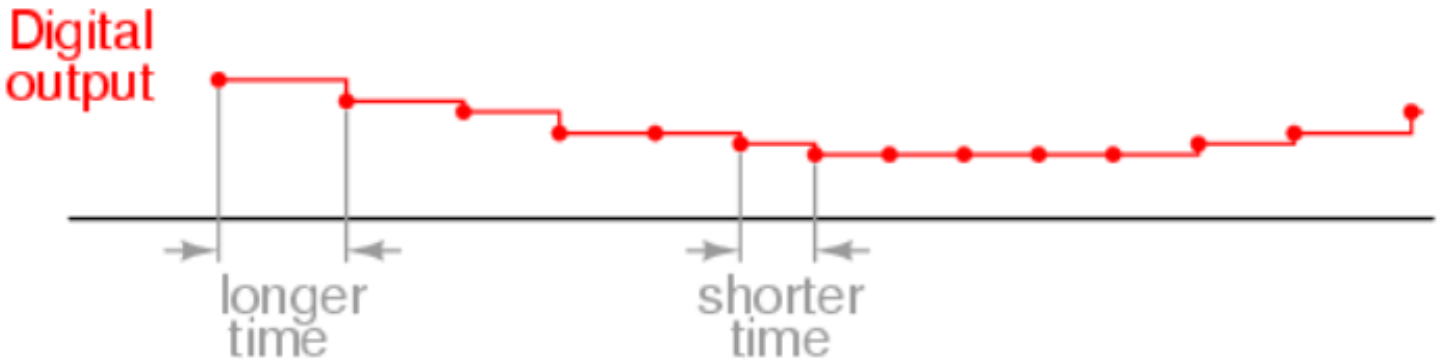


Source: http://dev.emcelettronica.com/analog-to-digital-converter-digital-to-analog-converter-overview

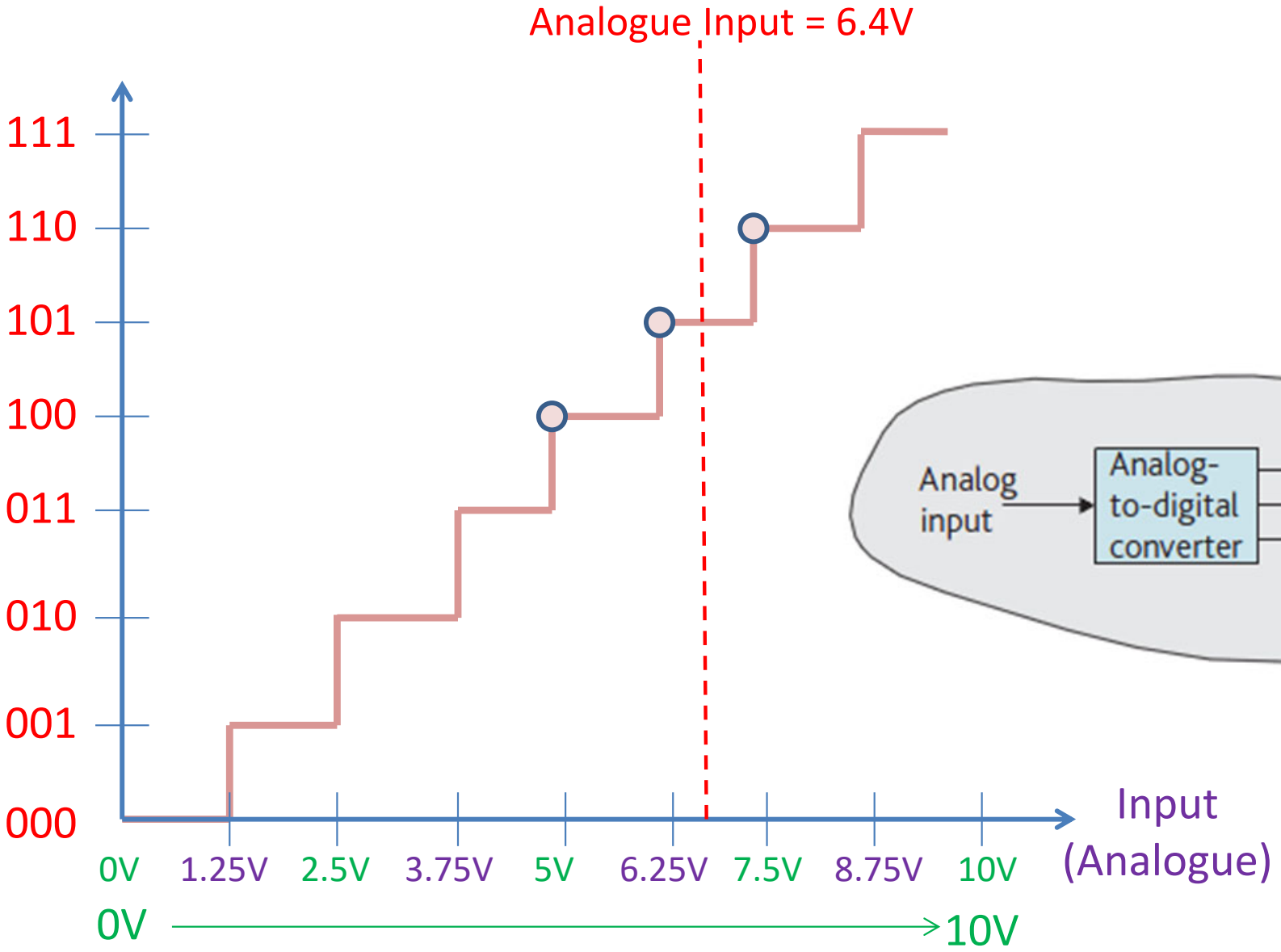
Digital Ramp ADC (Cont.)



- Disadvantage: Time between new digital output values changes depending on how high the input voltage is

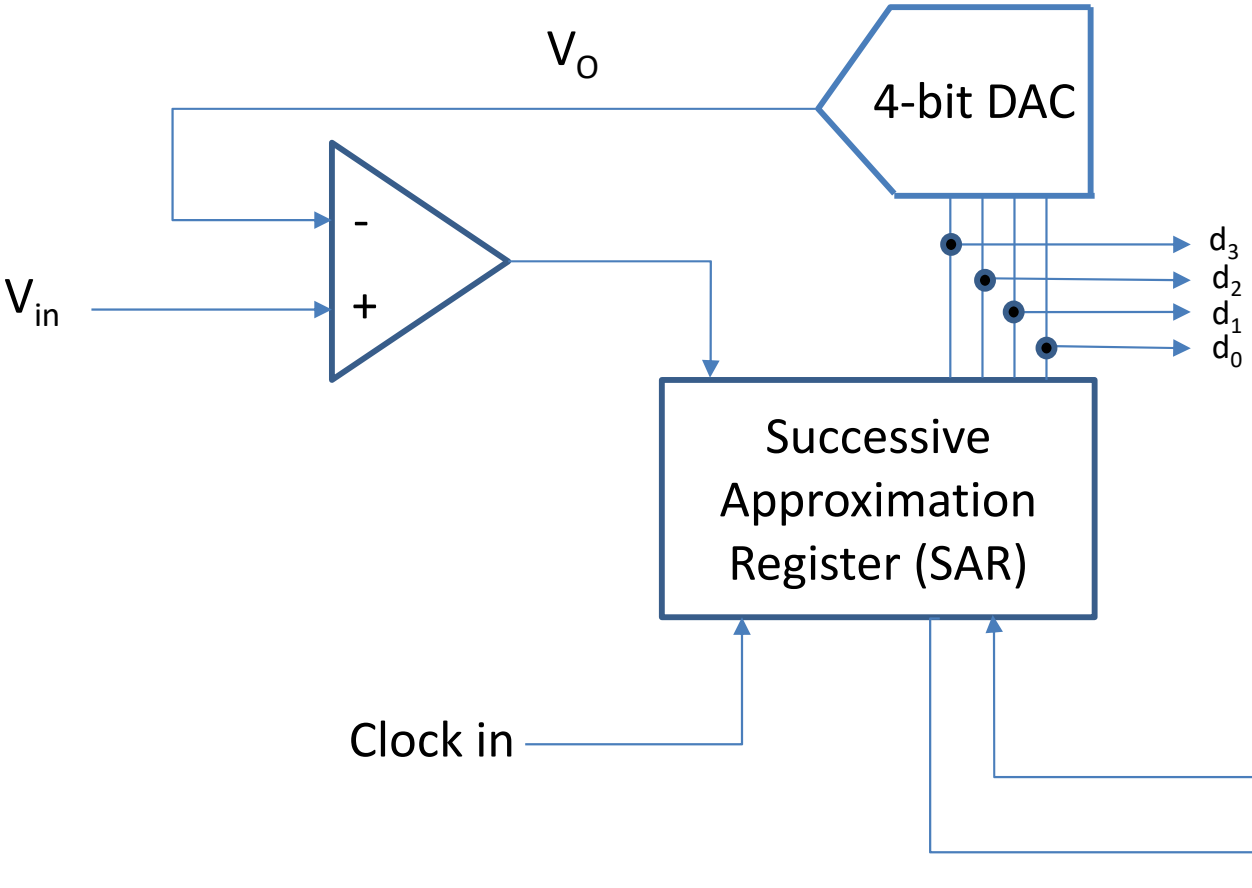


Input-Output Characteristic of an ADC

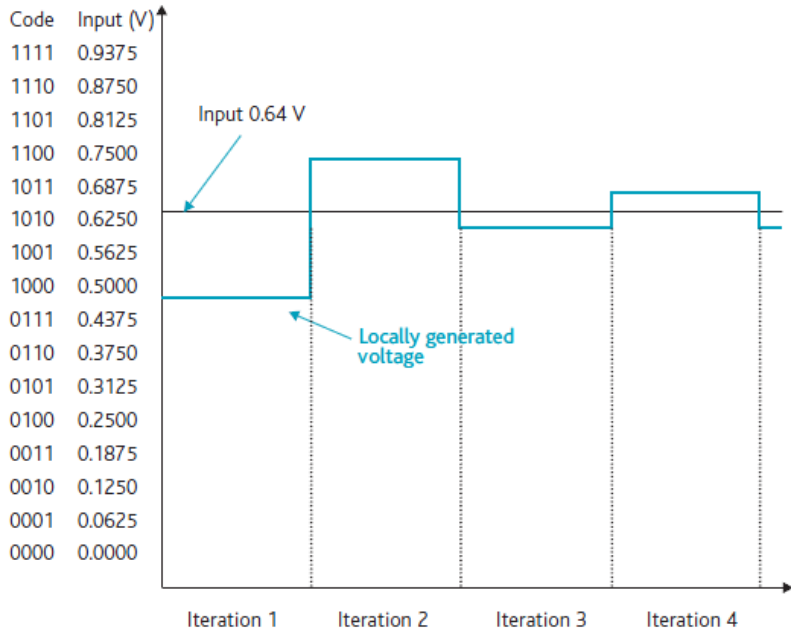


Successive Approximation

- Idea is to take large steps in increasing the locally generated analog signal (V_o) in the beginning, and reduce the step size as V_o converges to V_{in}

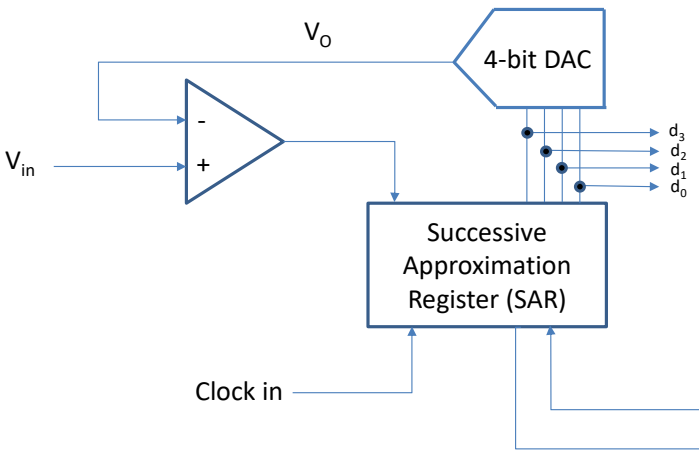


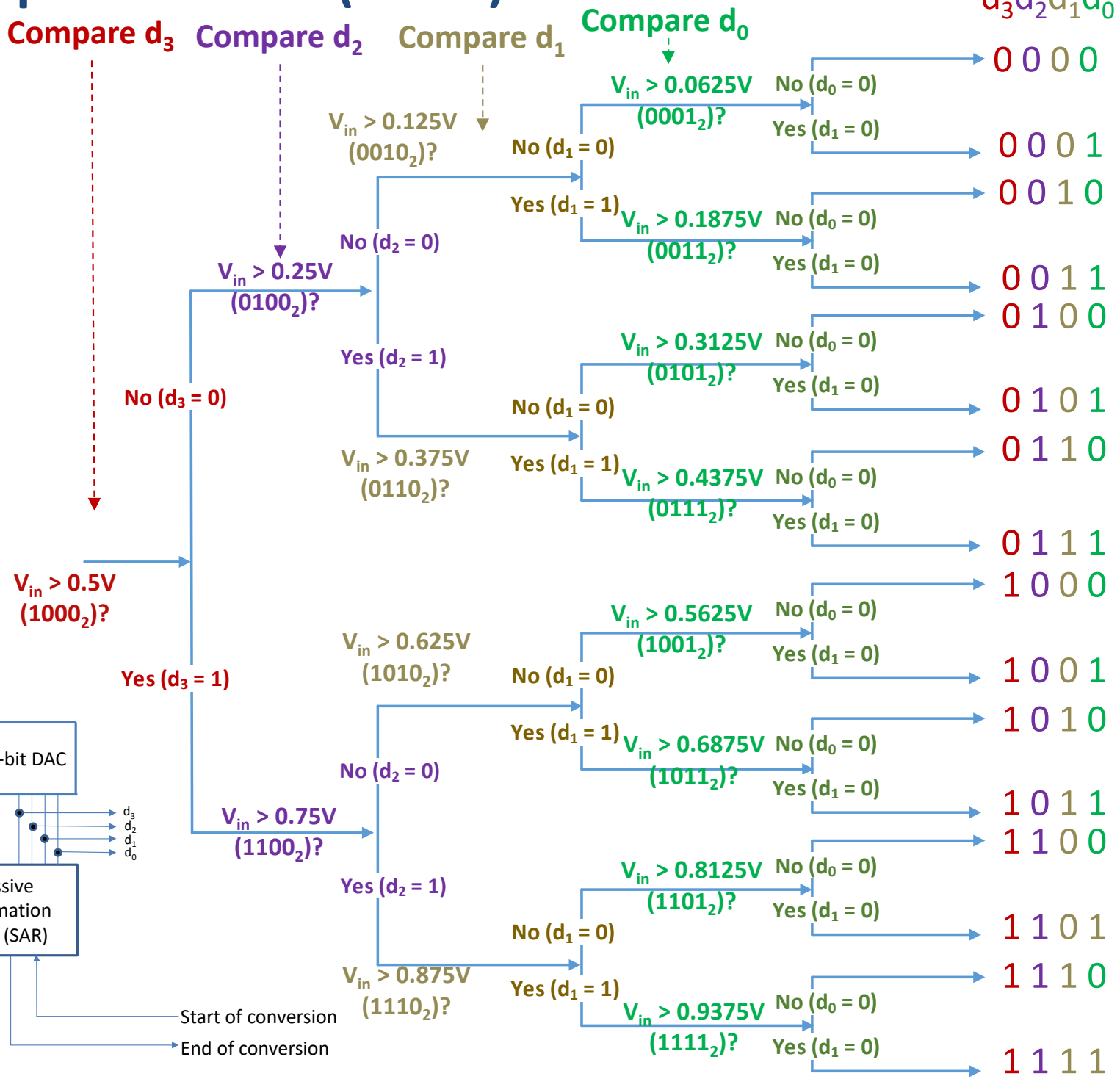
Code	Input (V)
1111	0.9375
1110	0.8750
1101	0.8125
1100	0.7500
1011	0.6875
1010	0.6250
1001	0.5625
1000	0.5000
0111	0.4375
0110	0.3750
0101	0.3125
0100	0.2500
0011	0.1875
0010	0.1250
0001	0.0625
0000	0.0000



Successive Approximation (Cont.)

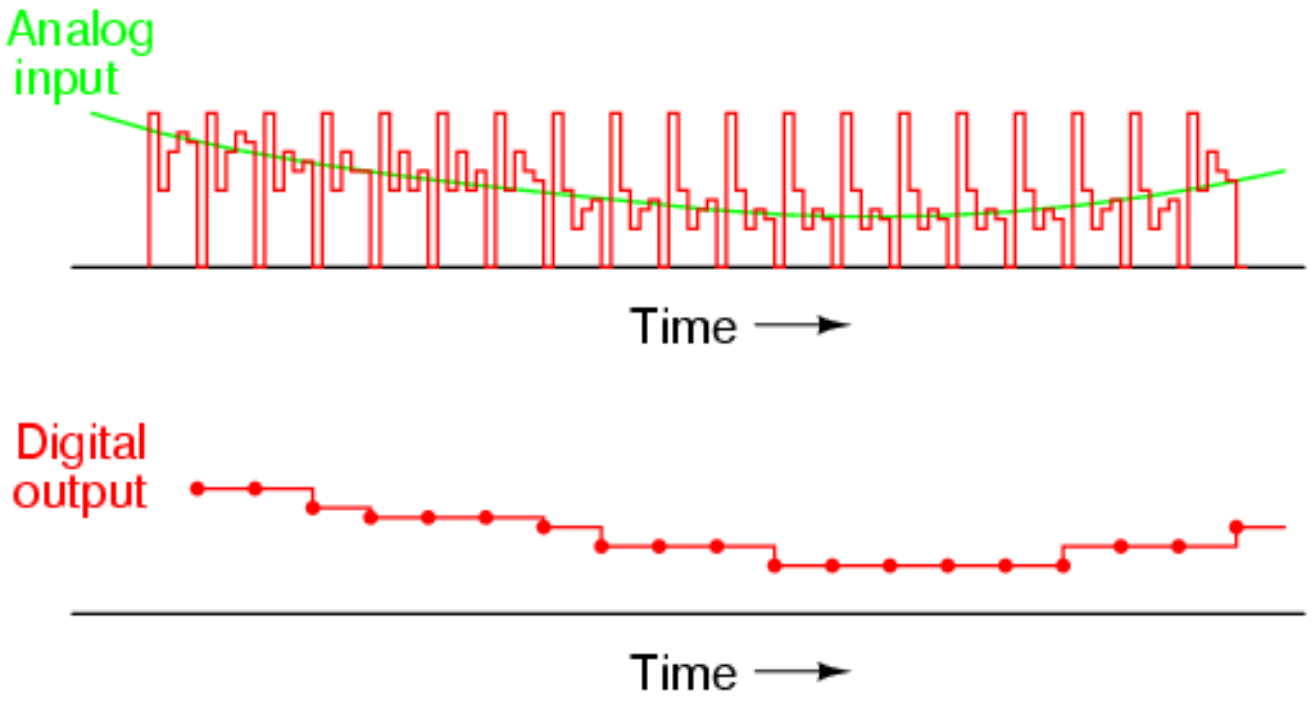
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0101	0.3125
0100	0.2500
0011	0.1875
0010	0.1250
0001	0.0625
0000	0.0000





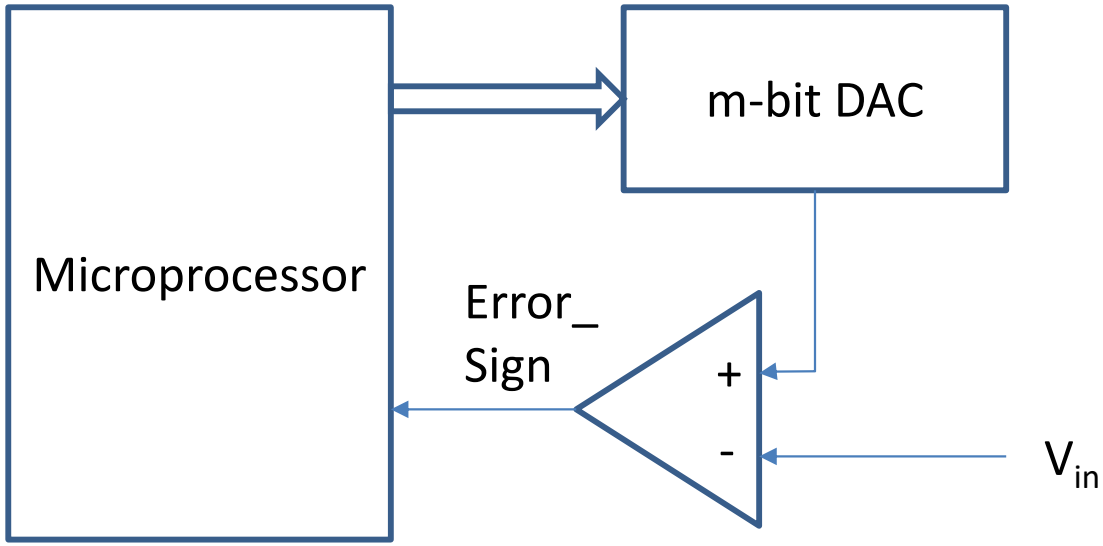
Successive Approximation (Cont.)

- Guarantees an m-bit conversion in no more than m iterations
- Produces faster results than the digital ramp ADC



- Can be implemented partially in software to reduce hardware cost

Successive Approximation (Cont.)




```
Successive Approximation
  DAC_output = 0
  Increment  = ½ Full-Scale Output (1000 … 000)
  For I = 1 to m
    DAC_output = DAC_output + Increment
    If Error_sign negative then
      DAC_output = DAC_output – Increment
    End If
    Increment = Increment/2
  End For
End Successive Approximation
```

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1 August 2022